

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Vanguard: Retail Revenue & Customer Decision Intelligence Platform
IBM Project Submission Track: Data Analytics & Applied AI

Project Title	Vanguard: Retail Decision Intelligence Platform
Domain / Track	IBM Applied Data Analytics & Business Intelligence
Repository	https://github.com/hs-techlabs/ibm-project.git
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Submission Status	Final Deliverable Release — Ready for Evaluation

1. Executive Summary & Purpose

This Software Requirements Specification (SRS) defines the structural, functional, and environment requirements for Vanguard, an enterprise-grade Decision Intelligence Platform built upon 533,878 verified transactions from the UCI Machine Learning Repository. Vanguard bridges raw transaction records and executive boardroom decisions by operationalizing the continuum: **Data** → **Information** → **Insights** → **Decisions** → **Actions**.

2. Functional Requirements (FR)

- FR-01: Data Ingestion & Sanitization:** Ingest 533k+ transactions, prune non-retail codes, isolate returns (InvoiceNo 'C'), reconcile guests, and cache via PyArrow.
- FR-02: Executive Decision Cockpit:** Display 6 primary C-suite KPIs (Net Revenue, Gross Profit, Total Orders, AOV, Gross Margin %, Return Rate %) with period deltas.
- FR-03: Revenue & Sales Velocity Module:** Render dual-axis monthly revenue versus order volume trends and intraday weekday-by-hour order density heatmap.
- FR-04: Catalog & Category Intelligence:** Map SKU titles into 8 departments via keyword taxonomy and compute cumulative Pareto 80/20 SKU distribution curves.
- FR-05: Algorithmic RFM Customer Segmentation:** Score accounts on Recency, Frequency, and Monetary quintiles (1-5), classifying Champions, Loyalists, and At-Risk cohorts.
- FR-06: Corporate Risk & Opportunity Engine:** Quantify domestic UK revenue exposure (82.4%) and size cross-border wholesale expansion vectors (\$485 international AOV).
- FR-07: Predictive Demand & Price Elasticity Simulator:** Execute Holt-Winters time-series forecasting with 95% CI and dynamic sliders for pricing and COGS inflation scenarios.
- FR-08: Consolidated Single-File Architecture:** Provide single-click execution via self-contained project.py adhering strictly to IBM submission standards.

3. Non-Functional Requirements (NFR)

- NFR-01: Performance & Startup Latency:** Sub-second deserialization and dashboard boot (< 2 seconds) via PyArrow Snappy Parquet caching.
- NFR-02: Submission Size Compliance:** Total packaged submission archive MUST strictly not exceed 10.0 MB for portal compatibility.
- NFR-03: Self-Healing Data Resilience:** Automatic fallback pipeline downloads, cleans, and generates cache if local files are missing.
- NFR-04: Cross-Platform Portability:** Identical, reproducible execution across macOS, Linux, and Windows 10/11 environments.

4. System Environment & Software Dependencies

Python	3.10+	Core runtime environment
Streamlit	1.30+	Executive web application interface
Plotly	5.18+	Interactive time-series, Pareto curves, and heatmaps
Pandas & NumPy	2.0+ / 1.24+	Vectorized ETL, RFM scoring, and aggregations
SciPy & Scikit-Learn	1.10+ / 1.3+	Statistical regressions and Holt-Winters forecasting
PyArrow & OpenPyXL	14.0+ / 3.1+	High-speed Parquet cache & Excel ingestion

5. Quick Verification & Execution

Evaluators can verify the application with: `git clone https://github.com/hs-techlabs/ibm-project.git` && `pip install -r requirements.txt` && `streamlit run project.py`